ModelIt!

Teacher's Guide

How Does Your Brain Know That's Hot?

The Lightning-Fast Science of Senses and Signals

Quick Reference

Grade Level	Grade 4
NGSS Standard	4-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-LS1-2

Use a model to describe that animals receive different types of information through their senses, process the information in their brain, and respond to the information in different ways.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how heat source intensity, nerve signal speed, and brain response time work together in the nervous system.
Disciplinary Core Idea	**LS1.D Information Processing** — Different sense receptors are specialized for particular kinds of information; animals use their perceptions and memories to guide their actions.
Crosscutting Concept	**Cause and Effect** — Students explore how the intensity of a stimulus (cause) affects the speed and strength of the nervous system's response (effect).

CAST Testing Alignment

MULTIPLE CHOICE:

A student accidentally touches a hot pan and pulls their hand back instantly, even before feeling pain. Which best explains why the hand moved before the brain processed the pain?

CONSTRUCTED RESPONSE:

Describe the journey of a signal from when your finger touches a hot object to when your hand pulls away. Use the words 'sensory receptor,' 'nerve,' 'signal,' and 'response' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Your nervous system is the most incredible communication network on Earth. It connects every part of your body to your brain, allowing you to sense the world, make decisions, and react to danger in fractions of a second. Understanding how signals travel through your body helps explain everything from reflexes to memory.

Sensory Receptors

Your body has millions of sensory receptors — specialized cells that detect specific types of information. Thermoreceptors in your skin detect temperature (hot and cold). Mechanoreceptors detect pressure and touch. Nociceptors detect pain. Photoreceptors in your eyes detect light. Chemoreceptors in your nose and tongue detect chemicals (smell and taste). Each type converts its specific stimulus into an electrical signal.

How Nerve Signals Travel

When a sensory receptor is activated, it generates an electrical signal called a nerve impulse. This impulse travels along nerve fibers (neurons) at speeds up to 120 meters per second — that is 270 miles per hour! The signal passes from one neuron to the next through tiny gaps called synapses. Signals from your finger can reach your brain in about 20 milliseconds.

Reflexes — The Shortcut

For extreme dangers like touching a hot stove, your body has a shortcut called a reflex arc. Instead of waiting for the signal to travel all the way to the brain and back, the signal goes only to the spinal cord, which immediately sends a response signal back to the muscles to MOVE. Your hand pulls away in about 50 milliseconds — before your brain even registers pain! This reflex system evolved because those fractions of a second can prevent serious injury.

The Brain — Mission Control

The brain is the command center of the nervous system. It receives signals from all sensory receptors, processes the information, stores memories, makes decisions, and sends signals back out to muscles and organs. The brain processes about 11 million bits of sensory information every second, but you are only aware of about 50 bits at a time. Most processing happens automatically without you thinking about it!

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify heat source intensity as an external component and nerve signal speed and brain response time as internal components that respond to the intensity of the stimulus.

E **Establish Relationships**
Students determine that heat source intensity positively affects nerve signal speed (hotter objects trigger faster signals), and faster nerve signals positively reduce brain response time.

V **Visualize & Model**
Students build a model showing how more intense heat triggers faster nerve signals and quicker brain responses, creating a survival system that reacts faster to greater danger.

E **Evaluate Outcomes**
Students run warm cup, hot stove, and boiling water scenarios to observe how increasing heat intensity affects signal speed and response time.

R **Revise & Extend**
Students add pain level or nerve pathway length to explore why some body parts are more sensitive than others.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Quick — clap your hands! How did your brain tell your hands to do that so fast? The signal traveled at over 200 miles per hour!"*

Watch For

Have students clap, then ask: How did the message get from your brain to your hands? Could you feel it traveling?

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to discover the incredible alarm system hidden inside your body."*

Watch For

Have students read objectives aloud. Pre-teach 'nerve' and 'signal' by comparing nerves to electrical wires.

Slide 3: Big Question

3 min

Say: *"Have you ever touched something really hot and your hand flew away on its own? You did not even think about it! How is that possible?"*

Watch For

Have students share stories of fast reflexes. Then ask: How does the message get from your finger to your muscles?

Slide 4: LEVER Framework

2 min

Say: *"We are going to build a model of your body's alarm system to understand how you react to danger."*

Watch For

Preview LEVER steps. Explain that we are modeling something invisible happening inside their bodies.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of your body's danger detection system?"*

Watch For

Guide sorting. Explain that heat source intensity is external because it comes from outside the body. Signal speed and response time are internal.

Slide 6: Activity 2: Connections

8 min

Say: *"When the heat is more intense, do you think the signal goes FASTER or SLOWER? Why would that help you survive?"*

Watch For

Help students draw positive arrows. Discuss why faster signals for bigger dangers makes sense for survival.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us test it! We will compare touching a warm cup vs. a hot stove vs. boiling water."*

Watch For

Students predict first. Run all three scenarios. The speed difference should surprise them!

Slide 8: Discoveries

5 min

Say: *"Your body is like a superhero — it reacts to danger even faster than you can think!"*

Watch For

Demonstrate the ruler-drop reflex test. Students test each other's reaction times and connect results to the model.

Slide 9: STEM Challenge

12 min

Say: *"Can you build a model that works like your nervous system? Detect, send, respond!"*

Watch For

Distribute materials (dominoes, string, bells, cups). Groups build alarm systems that detect a 'signal' and trigger a 'response.'

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Heat Source Intensity
- **INTERNAL:** Nerve Signal Speed, Brain Response Time

Relationships

- **Heat Source Intensity to Nerve Signal Speed:** POSITIVE (+) — A hotter, more dangerous stimulus triggers a stronger, faster nerve signal. Your body treats higher intensity as a bigger emergency, sending the alarm faster to protect you from injury.
- **Nerve Signal Speed to Brain Response Time:** POSITIVE (+) — Faster nerve signals mean the brain (or spinal cord for reflexes) receives the information sooner and can trigger a response more quickly. Speed is survival — the faster the signal, the faster you react.

Key Discoveries

- Your body has special sensor cells in your skin that detect heat, cold, pressure, and pain
- Nerve signals travel like electricity through wires — they can move over 100 meters per second through your body
- Stronger dangers (hotter objects) trigger faster, stronger nerve signals because your body treats it as an emergency
- Some responses (like pulling your hand from a hot stove) happen so fast that the signal does not even reach your brain first — it is called a reflex

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Your brain controls every single movement you make"	Many fast responses (reflexes) bypass the brain entirely! When you touch something dangerously hot, the signal goes to your spinal cord, which triggers the muscle response immediately. Your brain finds out about the pain AFTER your hand has already moved. This reflex system exists because waiting for the brain would be too slow for extreme dangers.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Nerves only send signals about pain"	Nerves carry ALL kinds of sensory information — not just pain! They send signals about temperature, pressure, texture, position, balance, sound, light, smell, and taste. You feel the softness of a pillow, the coldness of ice cream, and the tickle of a feather all through nerve signals. Pain is just one of many types of information nerves carry.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Signals travel instantly — there is no delay"	Nerve signals are fast but NOT instant. They travel at 1 to 120 meters per second depending on the nerve type. A signal from your toe to your brain takes about 60 milliseconds. That is why you can sometimes feel a stubbed toe in stages — you see it happen, then feel the pressure, THEN feel the pain. Each signal type travels at a different speed!